STAT 516 MIDTERM

1. (10pt) A certain item is produced in a factory on one of three machines, A, B, or C. The percentages of items produced on machines A, B, and C, are 50%; 30%; and 20% respectively. Moreover, 4% of A's products , 2% of B's products, and 4% of C's products, respectively, are defective. What is the percentage of all copies of this item that are defective?

   By the law of total probability we have

$$P(D) = P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C)$$
$$= (0.5)(0.04) + (0.3)(0.02) + (0.2)(.04) = 0.034$$

2. (10pt) Although many people buy lottery tickets out of an expectation of good luck, probabilistically speaking, buying lottery tickets is usually a waste of money. Indeed, suppose that in a weekly state lottery five of the numbers 0, 1, ..., 49 are selected without replacement at random and someone who holds exactly those numbers wins the lottery. Please compute the following (no need to do arithmetic calculations).

   (a) (5pt) If you hold one ticket, what is the probability that you're a winner in a given week?

   (b) (5pt) Now, assume that weekly lotteries are independent. If you keep buying a ticket every week for 40 years, what is the probability that you're going to win a lottery in at least one week?

   Then, the probability that someone holding one ticket will be the winner in a given week is $\frac{1}{\binom{50}{5}} = 4.72 * 10^{-7}$ Suppose this person buys a ticket every week for 40 years. Then, the probability that he will win the lottery in at least one week is $1 - (1 - 4.72 * 10^{-7})^{52*40} = 0.0098$ - still a very small probability. We assumed in this calculation that the weekly lotteries are all mutually independent, a reasonable assumption. The calculation would fall apart if we did not make this independence assumption.

3. (10pt)Take a random variable X that has discrete uniform distribution on $\{1, 2, \ldots, n\}$.

   (a) (5pt)Compute its moment generating function

   (b) (5pt)Using the moment generating function you just obtained, compute the expectation of X

   The moment generating function is

$$M(s) = \mathbb{E}e^{sX} = \sum_{x=1}^{n} e^{sx} P[X = x] = \frac{1}{n} \sum_{x=1}^{n} e^{xs} = \frac{e^s(e^{sn} - 1)}{n(e^s - 1)}$$

   The first derivative is

$$M'(s) = \frac{e^s(1 + ne^{(n+1)s} - (n+1)e^{ns})}{n(e^s - 1)^2}$$

   For $s = 0$, applying L'Hôpital's rule twice, we obtain $\mathbb{E}X = \frac{n+1}{2}$

4. (10pt) A lion sets a circular territory for itself by choosing a radius at random according to a standard exponential density (the unit being a mile). Compute the expected area of the lions territory

   Denote by X the radius of the lions territory; then X has the density e^{-x} if $x > 0$ and 0 otherwise. Since the area of a circle with radius x is πx^2, we have

$$\mathbb{E}(area) = \int_0^\infty \pi x^2 e^{-x}\, dx = \pi \int_0^\infty x^2 e^{-x}\, dx = 2\pi$$